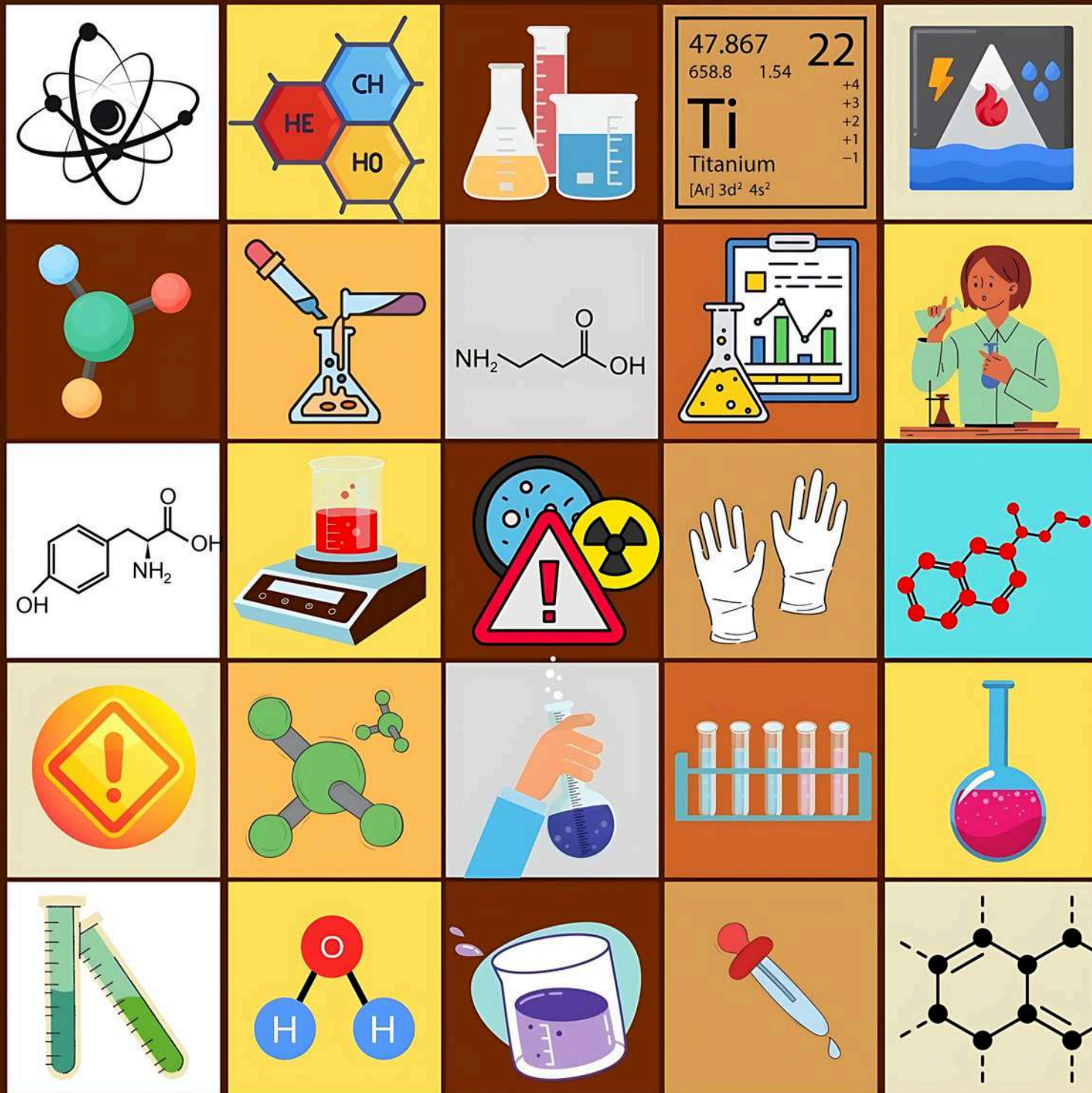




CLASS 9th NOTES CHEMSITRY

ATOMS AND MOLECULES

PRASHANT KIRAD

CLICK HERE TO BUY LATEST EDUCART BOOKS ON AMAZON - <https://amzn.to/4n3AKay>

Atoms and molecules

No. of Reactants = No. of Products

Laws of Chemical combination

- **Law of Conservation of Mass** - Mass can neither be created nor be destroyed.
- **Law of Constant properties** - In a chemical substance, the elements are always present in definite proportions by mass.

Dalton's atomic theory

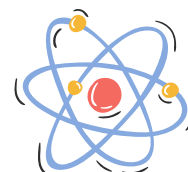
Atoms are:

- Tiny particle
- Indivisible
- Identical mass and chemical properties
- Different mass and chemical properties
- Combine in the same ratio



What is atom?

- The smallest particle of an element.
- Each atom shows all the properties of the element.



Atomic radius is measured in nanometers.

$$\begin{aligned} 1/10^9 &= 1 \text{ nm} \\ 1\text{m} &= 10^9 \text{ nm} \end{aligned}$$

→ No = नौ = 9

IUPAC (International Union of Pure and Applied Chemistry) approves names of elements, symbols, and units.

For example - Hydrogen(H), Aluminum(Al), Sodium(Na), etc.

Some elements have symbols according to the first letter of the name and a letter appearing later in the name. *E.g., Chlorine (Cl), Zinc(Zn), etc.*

Other symbols have been taken from the names of elements in Latin, German or Greek. *E.g., Symbol of Iron is Fe from its Latin name Ferrum and symbol of potassium is K from Kalium.*

Atomic mass

Mass of an atom equals to $\frac{1}{12}$ of mass of C-12 atom.

$$1 \text{ amu or } u = \frac{1}{12} \times \text{Mass of an atom of C}$$

$$1 u = 1.66 \times 10^{-27} \text{ kg}$$



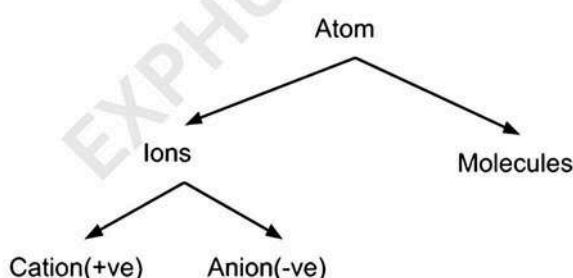
mostly double krdo
atomic number ko

Atomic Mass of some elements

Element	Symbol	Atomic Mass	Element	Symbol	Atomic Mass
Hydrogen	H	1u	Sodium	Na	23u
Helium	He	4u	Magnesium	Mg	24u
Lithium	Li	7u	Aluminium	Al	27u
Beryllium	Be	9u	Silicon	Si	28u
Boron	B	11u	Phosphorous	P	31u
Carbon	C	12u	Sulphur	S	32u
Nitrogen	N	14u	Chlorine	Cl	35u
Oxygen	O	16u	Potassium	K	39u
Fluorine	F	19u	Calcium	Ca	40u
Neon	Ne	20u	Iron	Fe	56

How do Atoms exist?

- Atoms form of most elements is not able to exist independently.
- Atoms form molecules and ions.



What is a molecule?

Group of two or more atoms that are chemically bonded together, that is tightly held together by attractive forces.

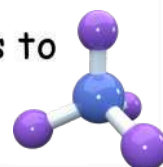
E.g., A molecule of oxygen consists of two atoms of oxygen and hence it is known as a diatomic molecule, O_2 .

The number of atoms constituting a molecule is known as its **atomicity**.

Helium is monoatomic, Ozone(O_3) is triatomic, Phosphorus (P_4) is tetraatomic.

Molecules of compounds

Atoms of different elements join together in definite proportions to form molecules of compounds.



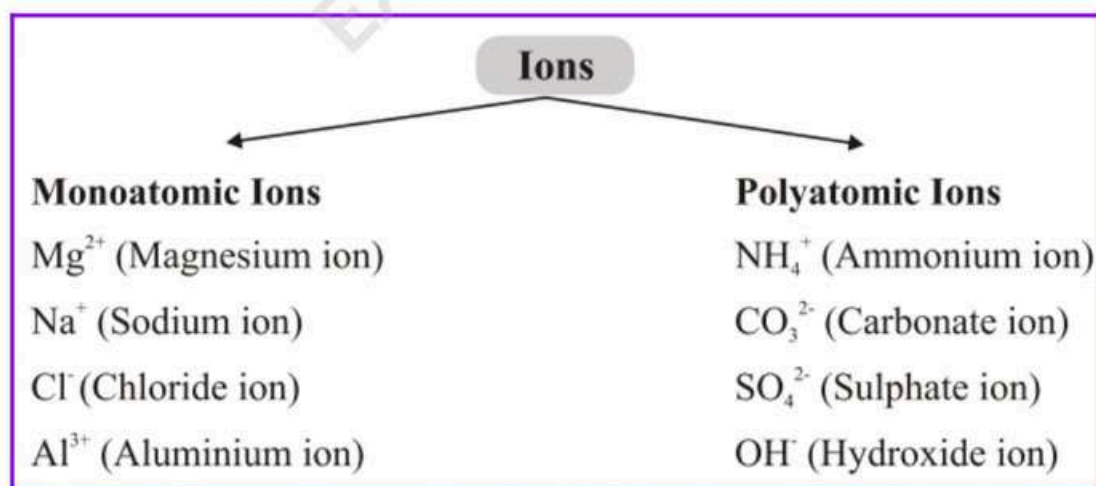
Compound	Combining elements	Ratio by mass
Water (H_2O)	Hydrogen, Oxygen	1:8
Ammonia (NH_3)	Nitrogen, Oxygen	14:3
Carbon dioxide (CO_2)	Carbon, Oxygen	3:8

What is an ion?

Compounds composed of metals and non-metals contain charged species.

The charged species are of two types:

- Anion - Negatively charged ion
- Cation - Positively charged ion



C⁺a⁺tion

a = addition = positive

A⁻n⁻ion

n = negative



*Ab Confuse matt hona!
- "Prashant Bhaiya"*

Chemical formulae

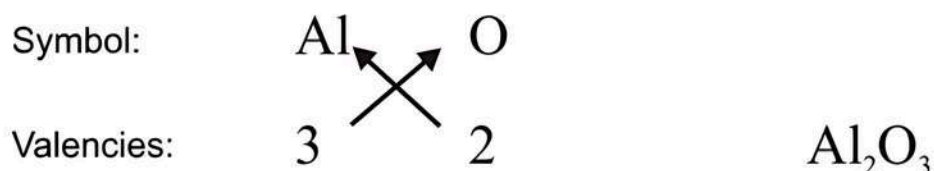
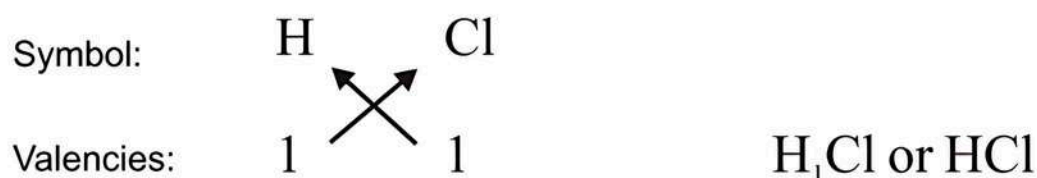
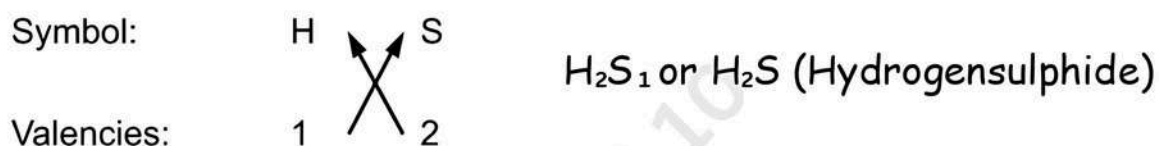
Symbolic representation of the composition of a compound.

Characteristics of chemical formulae:

- The valencies or charges on ion must balance.
- When a compound is formed of metal and non-metal, symbol of metals comes first. E.g., CaO, NaCl, CuO.
- When polyatomic ions are used, the ions are enclosed in brackets before writing to show that ratio. E.g., $\text{Ca}(\text{OH})_2$, $(\text{NH}_4)_2\text{SO}_4$

Examples:

Important!!!



Acche se practice krna!!
10th me bhi kaam aayega



Chemical formulae

For ions:

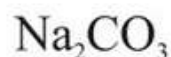


Class 9th Phodenge
- Prashant Bhaiya

Sodium carbonate

Symbol: Na CO₃

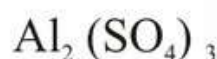
Valencies: 1+ 2-



Aluminium sulphate

Symbol: Al SO₄

Valencies: 3+ 2-



Molecular mass

The sum of atomic masses of all the atoms in a molecule of that substance. E.g., Molecular mass of H₂O = 2 × Atomic mass of Hydrogen + 1 × Atomic mass of Oxygen.

Formula unit mass

The sum of the atomic mass of ions and atoms present in formula for a compound.

In NaCl, Na = 23 amu, Cl = 35.5 amu

So, Formula unit mass = 1 × 23 + 1 × 35.5 = 59.5 u

Molar mass

The molar mass of a substance is the mass of 1 mole of that substance. It is equal to the 6.022×10^{23} atoms of that element/substance.

E.g., Atomic mass of hydrogen is 1 and its molar mass is 1 g/mol

Important formulae

- Number of moles (n) = Given mass/Molar mass = m/M
- Percentage of any atom in given compound = (Mass of element × 100)/Mass of compound